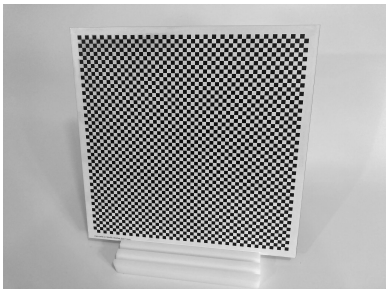
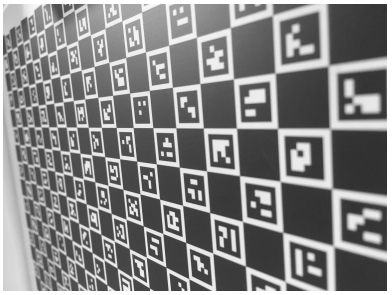
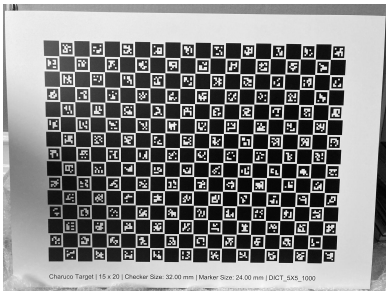
Lithographic Print Calibration Targets

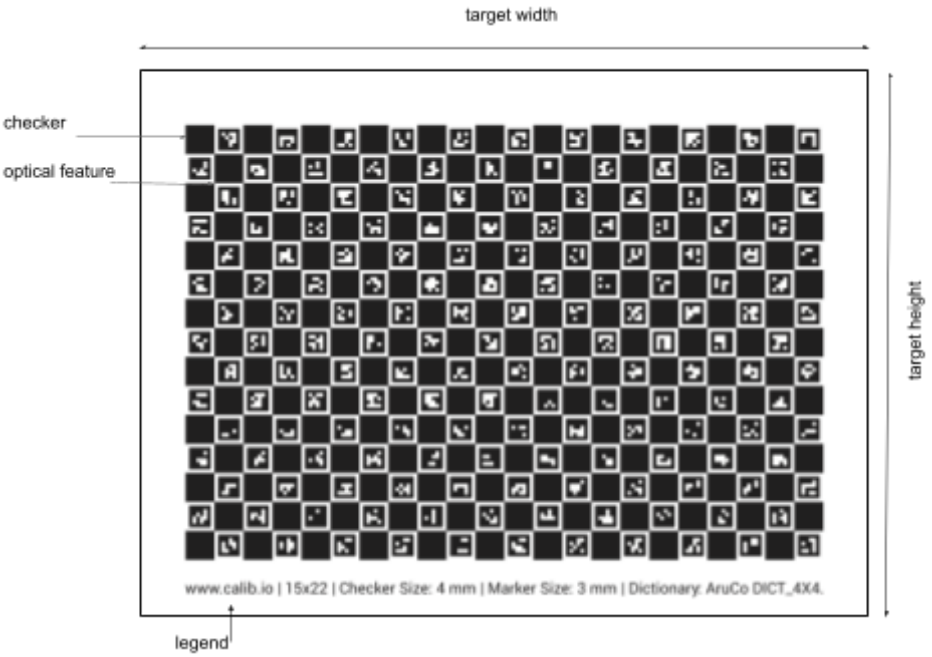




Optical Specifications

Patterning Process	Lithographic print patterning
Feature Distance Tolerance	0.04mm + 0.08mm/m
Flatness	0.1mm per 100mm (glass) / 0.5mm per 100mm (CFRP)
Resolution	10000 dpi
Gloss (@60deg)	2-4 (black) / 2 (white)
Opaqueness	85% (glass) / 100% (CFRP)
Illumination (recommended)	260 - 1400nm frontlit or backlit (glass)

Mechanical Specifications

Maximum Dimensions	800 x 600mm
Weight	16.64 kg/m² at 6mm thickness (default)
Temperature Resistance	5 °C to +60 °C [41 °F to +140 °F]
Specified temperature	20 °C (ISO 1:2016)
Linear thermal expansion	8 µm /(°C · m) (glass) / -0.8 µm /(°C · m) (CFRP)



Note: pattern position on target is not toleranced.

Substrate Properties

Lithographic print offers high feature accuracy and a matte finish for small to medium size calibration targets. Also suitable for active sensor calibration (i.e. structured light) due to low gloss and high opaqueness.

Lapped optical glass and carbon fiber reinforced polymer (CFRP) are offered as substrate options, both offering low thermal expansion. Lapped glass, while more fragile, has the best flatness, and can be used both as a frontlit and backlit target.

Custom patterns incur startup cost and approximately two weeks turnaround time.

For applications requiring larger targets, we recommend Alu composite targets with UV precision print. For the highest dimensional accuracy, ceramic etch lithographic targets are offered.

Dimensional tolerances

Ceramic lithography targets are manufactured on high end lithographic machines in a clean room environment. The manufacturing tolerance on this process is 0.04mm + 0.08mm/m, i.e. for two optical features spaced 1m apart, the maximum permissible distance error is 0.12mm.

The ISO 1101 flatness is best for the lapped glass substrate.

Customization

Lithographic targets are manufactured to customer requirements. Lead times are generally between 2 and 3 weeks.

- Custom patterns:
  - Minimum 1mm margin to target edges.
  - Provided as 100% scale PDF complete vector graphics.
- Customized legend text and/or logos.
- Drawing services can be offered. Ex:
  - ChAruCo targets with different dictionaries and/or marker ids.
  - Circle/checkerboard targets with unique finder patterns
  - Deltille grid patterns and other less common targets.
  - Single AruCo marker plates for VR and robot calibration purposes.
  - Resolution test charts.

Inspection and certification

Calibration targets are carefully inspected before shipping.

The following additional certifications can be purchased:

1. Calib.io Certification. We perform several distance and flatness measurements and certify that our manufacturing and product specifications are being met.
2. Measurement Report: The absolute 3D position of each saddle point is measured and provided in a measurement report and as a digital file. For targets up to 400mm, a calibrated optical coordinate measurement machine (OMM) is used in an ISO-17025 accredited measurement laboratory. For larger targets, we use a custom measurement setup.

Please contact us for a quote for these certification options.